ELEC 221 Lecture 15
Analysis of CT systems based on differential equations

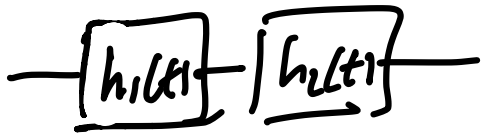
Thursday 29 October 2026

Announcements

- Midterm Monday during tutorial section
- Office hours tomorrow at 10am (instead of 12pm)

Last time: clarification

Correction: solution to quiz question.



$$H(j\omega) = \frac{1}{2+j\omega}\frac{1}{5+j\omega}$$

$$x(t) = \cos(3t) = \frac{1}{2}\left[e^{3jt} + e^{-3jt}\right]$$

~~$y(t) = H(j3)\,x(t)$~~

$$y(t) = H(3j)\cdot\frac{1}{2}e^{3jt} + H(-3j)\cdot\frac{1}{2}e^{-3jt}$$

$\vdots$

Last time

We analyzed the impulse and step response of an LTI system described by a first-order differential equation.

$$T\frac{dy(t)}{dt} + y(t) = x(t),$$

$$h(t) = \frac{1}{T} e^{-t/T} u(t)$$

$$s(t) = h(t) * u(t)$$

$$= (1 - e^{-t/T}) u(t)$$

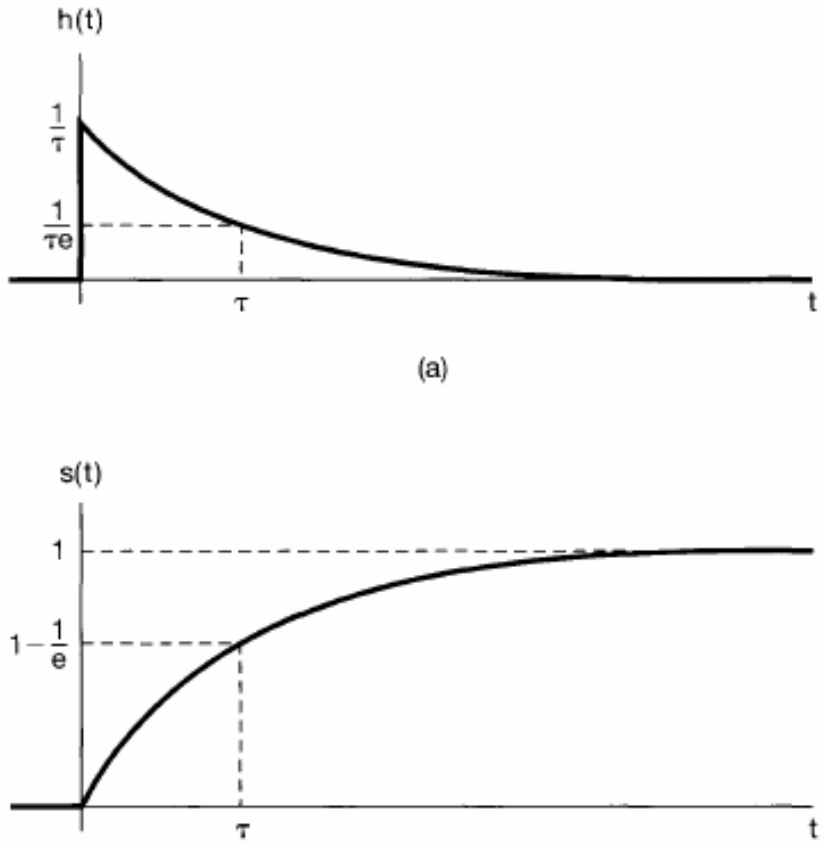


Image: Oppenheim Fig. 6.19

Last time

We derived the behaviour of the Fourier transform under differentiation and integration:

$$x(t) \longleftrightarrow X(j\omega)$$

$$\frac{dx(t)}{dt} \longleftrightarrow j\omega X(j\omega)$$

$$\int_{-\infty}^{t} x(\tau)\, d\tau \longleftrightarrow \frac{1}{j\omega} X(j\omega) + \pi X(0)\delta(\omega)$$

From these, we determined

$$\delta(t) \longleftrightarrow \Delta(j\omega) = 1$$

$$u(t) \longleftrightarrow U(j\omega) = \frac{1}{j\omega} + \pi\delta(\omega)$$

computes spectrum of $e^{-at}u(t)$ in lim $a \to 0$

Today

Learning outcomes:

- compute the Fourier spectrum of CT LTI systems described by differential equations
- characterize the oscillatory behaviour of CT systems described by second-order differential equations

Example: Fourier transform properties and differentiation

We can take advantage of differentiation and integration properties to simplify computations.

Suppose

$$x(t) \longleftrightarrow X(j\omega)$$

$$\frac{dx(t)}{dt} \longleftrightarrow j\omega X(j\omega)$$

What is the Fourier transform of

$$z(t) = \frac{d^2}{dt^2} x(t-1)$$

Two properties to take advantage of here:

- differentiation
- time shift: $x(t-t_0) \longleftrightarrow e^{-j\omega t_0} X(j\omega)$

Example: Fourier transform properties and differentiation

$$z(t) = \frac{d^2}{dt^2} x(t-1)$$

First, consider: $p(t) = x(t-1)$:

$$p(t) \longleftrightarrow P(j\omega)$$

$$\frac{d^2}{dt^2} p(t) \longleftrightarrow -\omega^2 P(j\omega) \quad \left((j\omega)^2 P(j\omega) \right)$$

$$P(j\omega) = X(j\omega) e^{-j\omega}$$

$$\Rightarrow Z(j\omega) = -\omega^2 e^{-j\omega} X(j\omega)$$

Fourier transforms and systems described by differential equations

Recall the LTI system described by a first-order ODE:

$$T\frac{dy(t)}{dt} + y(t) = x(t)$$

LTI systems can also be described by higher-order ODEs:

$$\sum_{k=0}^{N} \alpha_k \frac{d^k y(t)}{dt^k} = \sum_{k=0}^{M} \beta_k \frac{d^k x(t)}{dt^k}$$

We will derive a general expression for the frequency response of such systems.

Fourier transforms and systems described by differential equations

$$\sum_{k=0}^{N} \alpha_k \mathcal{F}\left(\frac{d^k y(t)}{dt^k}\right) = \sum_{k=0}^{M} \beta_k \mathcal{F}\left(\frac{d^k x(t)}{dt^k}\right)$$

$$\sum_{k=0}^{N} \alpha_k (j\omega)^k Y(j\omega) = \sum_{k=0}^{M} \beta_k (j\omega)^k X(j\omega)$$

$$Y(j\omega) \sum_{k=0}^{N} \alpha_k (j\omega)^k = X(j\omega) \sum_{k=0}^{M} \beta_k (j\omega)^k$$

$$H(j\omega) = \frac{Y(j\omega)}{X(j\omega)} = \frac{\sum_{k=0}^{M} \beta_k (j\omega)^k}{\sum_{k=0}^{N} \alpha_k (j\omega)^k}$$

Fourier transforms and systems described by differential equations

$$Y(j\omega)\sum_{k=0}^{N}\alpha_k(j\omega)^k = X(j\omega)\sum_{k=0}^{M}\beta_k(j\omega)^k$$

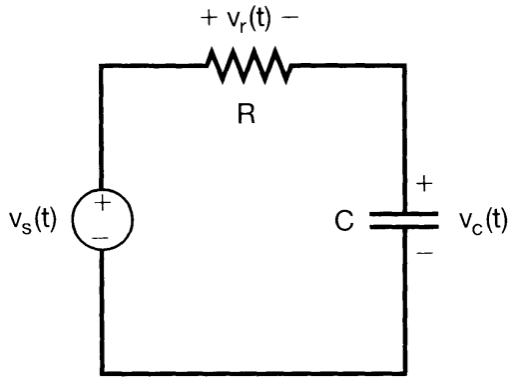
Final property:

$$H(j\omega) = \frac{Y(j\omega)}{X(j\omega)} = \frac{\sum_{k=0}^{M}\beta_k(j\omega)^k}{\sum_{k=0}^{N}\alpha_k(j\omega)^k}$$

This representation allows us to write down frequency response of systems described by ODEs **by inspection**! (and vice versa)

Exercise: frequency response of systems described by ODEs

What are the **impulse response** and **frequency response** of our RC circuit filter?



$$x(t) = v_s(t)$$
$$y(t) = v_c(t)$$

$$RC\frac{dy(t)}{dt} + y(t) = x(t)$$

$$H(j\omega) = \frac{Y(j\omega)}{X(j\omega)} = \frac{\sum_{k=0}^{M} \beta_k (j\omega)^k}{\sum_{k=0}^{N} \alpha_k (j\omega)^k} = \frac{1}{1 + j\omega RC}$$

Exercise: frequency response of systems described by ODEs

What are the **impulse response** and **frequency response** of the system described by

$$\frac{d^3 y(t)}{dt^3} - 4\frac{dy(t)}{dt} = 3\frac{d^2 x(t)}{dt^2} + x(t)$$

Start with frequency response:

$$H(j\omega) = \frac{Y(j\omega)}{X(j\omega)} = \frac{-3\omega^2 + 1}{-j\omega^3 - 4j\omega} = \frac{3(j\omega)^2 + 1}{(j\omega)^3 - 4j\omega}$$

Example: frequency response of systems described by ODEs

We can now leverage this to determine the impulse response:

$$H(j\omega) = \frac{3(j\omega)^2 + 1}{(j\omega)^3 - 4j\omega}$$

Use partial fractions:

$$H(j\omega) = \frac{3(j\omega)^2 + 1}{(j\omega)\left((j\omega)^2 - 4\right)}$$

$$= \frac{3(j\omega)^2 + 1}{(j\omega)(j\omega + 2)(j\omega - 2)}$$

$$= \frac{A}{j\omega} + \frac{B}{j\omega + 2} + \frac{C}{j\omega - 2}$$

$$\frac{3(j\omega)^2+1}{j\omega(j\omega+2)(j\omega-2)} = \frac{A}{j\omega} + \frac{B}{j\omega+2} + \frac{C}{j\omega-2}$$

$$= \frac{A(j\omega+2)(j\omega-2) + Bj\omega(j\omega-2) + Cj\omega(j\omega+2)}{j\omega(j\omega+2)(j\omega-2)}$$

$$= \underbrace{A\left((j\omega)^2-4\right) + B\left((j\omega)^2-2j\omega\right) + C\left((j\omega)^2+2j\omega\right)}_{\text{denom.}}$$

$$= (j\omega)^2\left[A+B+C\right] + (j\omega)\left[-2B+2C\right] - 4A$$

$$= 3(j\omega)^2+1$$

$$-\frac{1}{4} + 2B = 3 \Rightarrow B = \frac{13}{8}$$

$$\Rightarrow A = -1/4 \qquad B = C$$

Example: frequency response of systems described by ODEs

were not
Details ~~are~~ left as an exercise:

$$H(j\omega) = \frac{-1/4}{j\omega} + \frac{13/8}{j\omega+2} - \frac{13/8}{2-j\omega}$$

To get the impulse response, we can take the inverse Fourier transform, and leverage linearity:

$$e^{-at}u(t) \longleftrightarrow \frac{1}{a+j\omega} \qquad \mathrm{Re}(a) > 0$$

$$\frac{13/8}{j\omega+2} \rightarrow \frac{13}{8}e^{-2t}u(t) \qquad \frac{-13/8}{2-j\omega} \rightarrow -\frac{13}{8}e^{2t}u(-t)$$

Example: frequency response of systems described by ODEs

$$h(t) = -\frac{1}{4}\mathcal{F}^{-1}\left(\frac{1}{j\omega}\right) + \frac{13}{8}\mathcal{F}^{-1}\left(\frac{1}{j\omega+2}\right) + \frac{13}{8}\mathcal{F}^{-1}\left(\frac{1}{j\omega-2}\right)$$

Check Table 4.2 - two expressions to leverage:

$$\frac{1}{j\omega} = \underbrace{\frac{1}{j\omega} + \pi\delta(\omega)}_{u(t)} - \underbrace{\pi\delta(\omega)}_{1/2}$$

$e^{j2t} \leftrightarrow 2\pi\delta(\omega-2)$

So we have:

$$h(t) = -\frac{1}{4}u(t) + \frac{1}{8} + \frac{13}{8}e^{-2t}u(t) - \frac{13}{8}e^{2t}u(-t)$$

Second-order systems

Consider a system described by the ODE

$$\frac{d^2y(t)}{dt^2} + 2\zeta\omega_n\frac{dy(t)}{dt} + \omega_n^2 y(t) = \omega_n^2 x(t)$$

Exercise: what is the frequency response?

$$H(j\omega) = \frac{\omega_n^2}{(j\omega)^2 + 2\zeta\omega_n(j\omega) + \omega_n^2}$$

Second-order systems

Let's explore this in a little more detail and compute the impulse and step response of this system.

$$H(j\omega) = \frac{\omega_n^2}{(j\omega)^2 + 2\zeta\omega_n(j\omega) + 1}$$

$$= \frac{\omega_n^2}{(j\omega - c_+)(j\omega - c_-)}$$

where

$$c_\pm = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1}$$

Three cases to consider:

- $\zeta = 1$
- $\zeta > 1$
- $\zeta < 1$

Second-order systems

$$H(j\omega) = \frac{\omega_n^2}{(j\omega - c_+)(j\omega - c_-)}, \quad c_\pm = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1}$$

Case: $\zeta = 1$.

$$c_\pm = -\zeta\omega_n = -\omega_n$$

$$H(j\omega) = \frac{\omega_n^2}{(j\omega + \omega_n)^2}$$

Use handy table of Fourier transform pairs to find

$$h(t) = \omega_n^2 \, t \, e^{-\omega_n t} \, u(t)$$

$$s(t) = \left(1 - e^{-\omega_n t} - \omega_n t e^{-\omega_n t}\right) u(t)$$

Second-order systems

$$H(j\omega) = \frac{\omega_n^2}{(j\omega - c_+)(j\omega - c_-)}, \quad c_\pm = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1}$$

Case: $\zeta \neq 1$. Do a partial fraction expansion:

$$H(j\omega) = \frac{\omega_n}{2\sqrt{\zeta^2-1}} \frac{1}{j\omega - c_+} - \frac{\omega_n}{2\sqrt{\zeta^2-1}} \frac{1}{j\omega - c_-}$$

Use table of Fourier transform pairs to find

$$h(t) = \frac{\omega_n}{2\sqrt{\zeta^2-1}} \left(e^{c_+ t} - e^{c_- t} \right) u(t)$$

Second-order systems

$$h(t) = \frac{\omega_n}{2\sqrt{\zeta^2 - 1}} \left(e^{c_+ t} - e^{c_- t}\right) u(t), \qquad c_\pm = -\zeta\omega_n \pm \omega_n \sqrt{\zeta^2 - 1}$$

Exercise: derive the step response,

$$s(t) = \left[1 + \frac{\omega_n}{2\sqrt{\zeta^2 - 1}} \left(\frac{e^{c_+ t}}{c_+} - \frac{e^{c_- t}}{c_-}\right)\right] u(t)$$

The form of the exponential depends whether $\zeta > 1$ or $\zeta < 1$

- $\zeta < 1$: $c_\pm$ are imaginary; complex exponentials, so the response will oscillate!
- $\zeta > 1$: $c_\pm$ real and negative; decaying exponentials

Let's go plot these.

For next time

Content:

- Discrete-time FT
- New chapters: sampling; moodulation

Action items:

1. Prepare for midterm

Recommended reading:

- From today's class: Oppenheim 4.3.4, 6.5, 6.7
- Suggested problems: 4.33-4.36, 6.5, 6.9, 6.15, 6.18, 6.26, 6.27
- For next class: Oppenheim Ch. 5